SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Data Structures & Algorithms (Unit 1 to 5 Complete Notes)

Department: CSE	**Course / Subject:** Data Structures & Algorithms (CS301PC)
Semester: Semester 3	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Comprehensive reference notes on Stacks, Queues, Linked Lists, Trees (BST, AVL, B-Trees), Graph traversals (BFS/DFS), and Hashing techniques with complexity analysis.

Unit 1: Linear Data Structures

Abstract Data Types, Dynamic Arrays, Singly & Doubly Linked Lists, Circular Lists, Stack implementations, and Expression evaluation (Infix to Postfix).

Unit 2: Queues and Recursion

Linear Queue, Circular Queue, Double-Ended Queue (Deque), Priority Queue with Binary Heaps, and Recursion stack frame trace.

Unit 3: Non-Linear Structures: Trees

Binary Tree traversals, Binary Search Tree (BST) operations, Self-Balancing AVL trees with rotations (LL, RR, LR, RL), and Red-Black Trees.

Unit 4: Graph Algorithms

Graph representation (Adjacency Matrix & List), Depth First Search (DFS), Breadth First Search (BFS), Minimum Spanning Tree (Prim's & Kruskal's), and Dijkstra's Shortest Path.

Unit 5: Sorting & Hashing

Merge Sort, Quick Sort, Radix Sort, Open Addressing (Linear Probing, Quadratic, Double Hashing), Separate Chaining, and Collision Resolution.